Hands-on Exercise: Build an AI Agent with Python Tools

Duration: 90 Minutes
Technology: Python, PyCharm, OpenAI API (GPT-4o-mini)
Difficulty: Beginner

Problem Statement

Build an **AI Library Assistant** that understands user requests and decides which Python tool to execute.

The agent should be able to search for books and reserve available books for users.

Task 1: Create the Project

Create a Python project with the following structure:

```
Library_AI_Agent/
├── agent.py
├── tools.py
├── books.json
├── reservations.json
├── requirements.txt
└── .env
```

Task 2: Prepare Sample Data

Create `books.json` containing at least five books.

Each book should have:

- Book ID
- Title
- Author
- Category
- Available copies

Example:

```
{
  "id": 101,
  "title": "Python Crash Course",
  "author": "Eric Matthes",
  "category": "Programming",
  "available_copies": 3
}
```

Task 3: Build Two Python Tools

Tool 1: `search_books()`

- Accept a book title or keyword.
- Search the JSON data.
- Return matching books and their availability.

Tool 2: `reserve_book()`

- Accept a book ID and customer name.
- Check whether a copy is available.
- Ask for confirmation before creating the reservation.
- Save the reservation in `reservations.json`.
- Reduce the available copy count.
- Return the reservation details.

Do not create a reservation when no copies are available.

Task 4: Create the AI Agent

Using GPT-4o-mini and OpenAI function calling:

1. Accept questions from the user.
2. Provide the two tools to the AI model.
3. Let the model decide which tool to call.
4. Execute the selected Python function.
5. Send the tool result back to the model.
6. Display the final AI response.

Print the selected tool name in the terminal so the user can observe the agent's decision.

Task 5: Test the Agent

Test the following scenarios:

1. "Find books related to Python programming."
2. "Is Python Crash Course available?"
3. "Reserve Python Crash Course for Ahmed."
4. "Show me books related to artificial intelligence."
5. "Reserve a book that has no available copies."

Verify that the agent selects the appropriate tool and that reservations are saved only when approved.

Expected Outcome

The completed application must demonstrate:

- An AI model selecting between two Python tools.
- Execution of the selected tool.
- Reading and updating JSON data.
- Returning tool results to the AI model.
- Generating a final response based on the tool output.

Note: Use any AI Model of your choice even though the documet says GPT